Impact energy and TNT equivalents

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Convert an energy estimate without claiming a complete damage prediction.

SUCCESS CRITERIA - I can...

1. Convert speed units before squaring.
2. Calculate kinetic energy from stated model values.
3. Distinguish an energy equivalent from an explosion or damage forecast.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “equivalent” and “kinetic energy” to explain your model or result. Sentence starters are optional.

equivalent: equal in the specified quantity, not necessarily in other effects

kinetic energy: energy associated with motion

impact: a collision or strike between objects

01 Warm-up: bring back an earlier idea

Two students perform the same lift. One takes 3 times as long as the other.
Compare work and average power.

Word help: power = rate of energy transfer

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $K = \frac{1}{2}mv^2$; 1 ton TNT equivalent = 4.184×10^9 J. Equivalent energy does not mean identical effects.

Worked reasoning: A fictional event transfers 8.4×10^6 J. Using 4.2×10^6 J per kg TNT as an energy comparison gives 2 kg TNT equivalent. This compares energy, not identical effects.

Your turn: Draw a labeled energy comparison and name one effect this number alone cannot predict.

Sentence starter: I predict ... . My model supports this because ... .

Initial	Transfers	Final

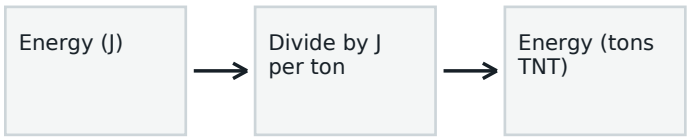
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two reports describe the same fictional impact: one uses joules and the other uses TNT-equivalent tons.
Use the conversion model to explain why the units can express the same quantity. What does this comparison leave out?

Sentence starter: My claim is ... . The diagram/data show ..., so ... .



Marzano check - circle: ① ② ③

04 Apply the physics

A supplied energy is 8.368×10^9 J. Divide by 4.184×10^9 J/ton.
Find TNT equivalent.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Energy (J)	J per ton TNT	Equivalent tons
___	4.184×10^9	___

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why should an energy-equivalent number not be reported as a precise area of destruction?

Sentence starter: My claim is … . The diagram/data show …, so … .

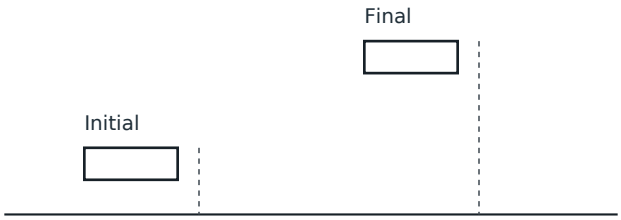
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Lift a 5 kg object by 7 m.
Use g=10 N/kg.
Find the change in gravitational potential energy.
Label the distances or energy levels used in your reasoning.

Word help: potential energy = energy associated with the configuration of interacting objects

Sentence starter: I found … [unit]. This value means … in this situation.



Choose and label a zero-height level.

Marzano check - circle: 1 2 3

07 Exit check

At fixed mass, speed halves. What happens to K and its TNT equivalent?
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .



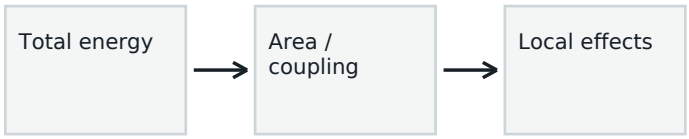
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two events have equal energy but spread it over different areas.
Explain why local effects could differ without changing total energy.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3